

Mathematical tables may be used in any question. The use of non-programmable, silent and cordless calculator is allowed.

Each question is followed by four options lettered A to D. Find the correct option for each question and shade in pencil, on your answer sheet, the answer space which bears the same letter as the option you have chosen.

Give only one answer to each question. An example is given below.

The ages, in years, of four boys are 10, 12, 14 and 18. What is the average age of the boys?

- A. 12 years
- B. $12\frac{1}{2}$ years
- C. 13 years
- D. $13\frac{1}{2}$ years

The correct answer is $13\frac{1}{2}$ years, which is lettered D, and therefore answer space D would be shaded. ☐ A ☐ B ☐ C ☒ D

Think carefully before you shade the answer spaces; erase completely any answers you wish to change.

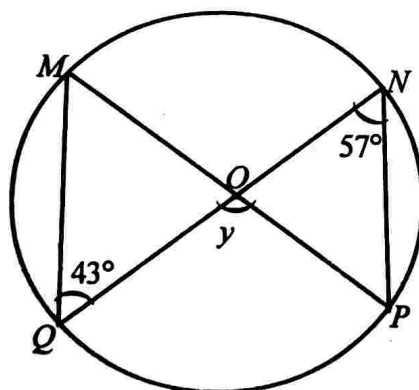
Do all rough work on this question paper.

Now answer the following questions.

1. If $\frac{1}{2}$ and -3 are the roots of $px^2 + qx + r = 0$, find the values of p , q and r .

- A. $p = -2$, $q = 5$, $r = 3$
- B. $p = 2$, $q = 5$, $r = -3$
- C. $p = 2$, $q = -5$, $r = 3$
- D. $p = 2$, $q = 5$, $r = 3$

2.



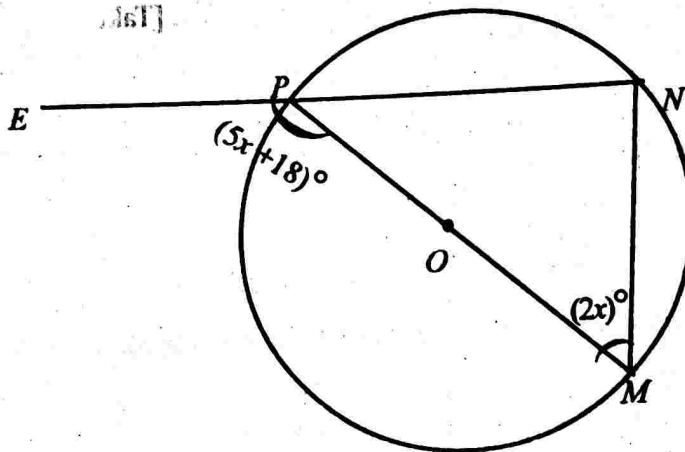
NOT DRAWN TO SCALE

In the diagram, $MNPQ$ is a circle, centre O . $\angle MQN = 43^\circ$ and $\angle QNP = 57^\circ$. Find the value of y .

- A. 100°
- B. 80°
- C. 90°
- D. 70°

3. If $\frac{4m+3n}{4m-3n} = \frac{5}{2}$, find the ratio of $m : n$.
- A. 3 : 4
B. 4 : 3
C. 7 : 4
D. 4 : 7
4. When the point (4, 5) is rotated through an angle in the anticlockwise direction about the origin, its image is (-5, 4). What is the angle of rotation?
- A. 90°
B. 300°
C. 180°
D. 270°
5. The ratio of girls to boys in a certain committee is 5 : 2. If there are 35 members in the committee, how many more boys must be added to the committee to have the ratio of girls to boys as 5 : 4?
- A. 10
B. 20
C. 15
D. 5
6. A box contains 2 red, 6 white and 5 black balls, all of the same size. If a ball is selected at random, what is the probability that it is black?
- A. $\frac{2}{13}$
B. $\frac{5}{13}$
C. $\frac{11}{13}$
D. $\frac{5}{11}$
7. Given that $p^2 + q^2 + r^2 = 50$, $p = 5$ and $\sqrt{q} = 2$, find the positive value of r .
- A. 4
B. 3
C. 2
D. 5

8.

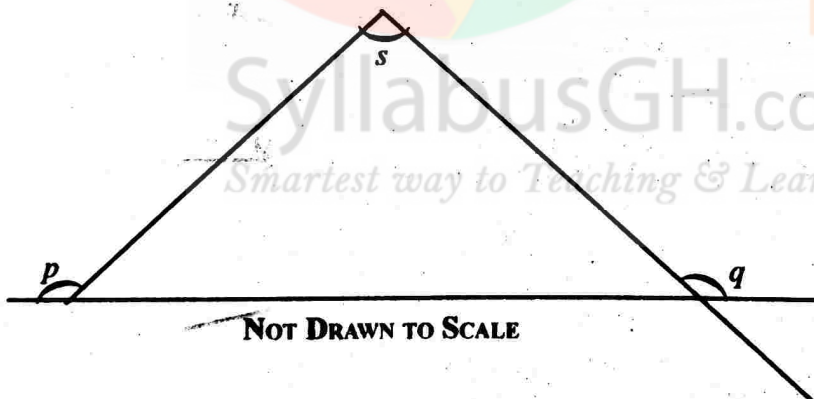


NOT DRAWN TO SCALE

In the diagram, MOP is the diameter of the circle MNP centre O . $\angle NMP = (2x)^\circ$ and $\angle EPM = (5x+18)^\circ$. Find the value of x .

- A. 72°
- B. 18°
- C. 48°
- D. 24°

9.



NOT DRAWN TO SCALE

In the diagram, $p + q = 250^\circ$. Find the angle marked s .

- A. 110°
- B. 70°
- C. 290°
- D. 250°

10. A sector which subtends an angle 150° is cut from a circular plate of radius 14 cm. Find, correct to one decimal place, the perimeter of the remaining plate. [Take $\pi = \frac{22}{7}$]

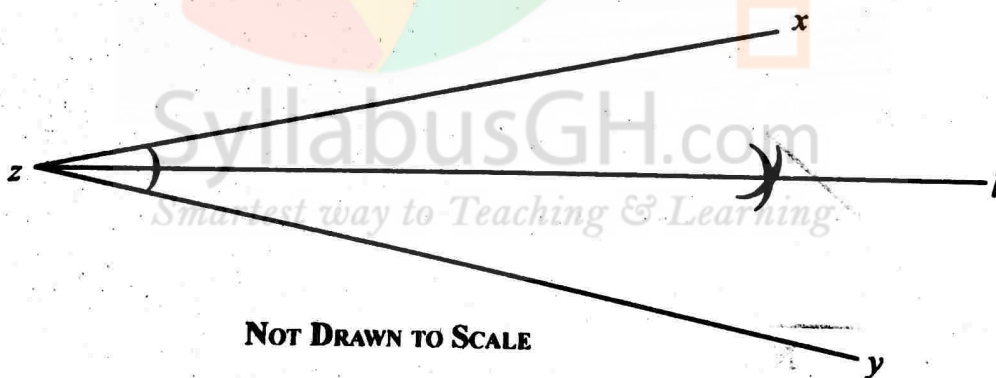
- A. 65.3 cm
- B. 79.3 cm
- C. 84.7 cm
- D. 64.7 cm

11. A rectangular tank of sides 4 m by 8 m by 11 m has the same volume as a cylindrical tank of height 7 m. Calculate the base radius of the cylindrical tank. [Take $\pi = \frac{22}{7}$]
- A. 22 m
B. 8 m
C. 4 m
D. 88 m

12. Abudu can do a piece of work in 6 days and Efah can do the same work in 3 days. What fraction of the work can both do together in a day?

- A. $\frac{1}{8}$
B. $\frac{2}{3}$
C. $\frac{1}{2}$
D. $\frac{1}{3}$

13.



Describe the locus, l in the diagram.

- A. Locus of points equidistant from x and y .
B. Locus of points equidistant from x and z .
C. Locus of points equidistant from xy and zy .
D. Locus of points equidistant from zx and zy .
14. The base radius and slant height of a solid cone are 8 cm and 14 cm respectively. Calculate, correct to two decimal places, its volume. [Take $\pi = \frac{22}{7}$]
- A. 838.67 cm^3
B. 553.14 cm^3
C. 640.87 cm^3
D. 770.32 cm^3

15. In a hall, there are 175 persons. 12 % are children, 56 are men and the rest are women. If one person is selected at random from the hall, find the probability that a woman is selected.

- A. $\frac{8}{25}$
- B. $\frac{3}{25}$
- C. $\frac{14}{25}$
- D. $\frac{11}{25}$

16. Given that $\tan x = \frac{12}{5}$, find the value of $(\sin x \cos x)$.

- A. $\frac{165}{60}$
- B. $\frac{60}{169}$
- C. $\frac{169}{65}$
- D. $\frac{65}{169}$



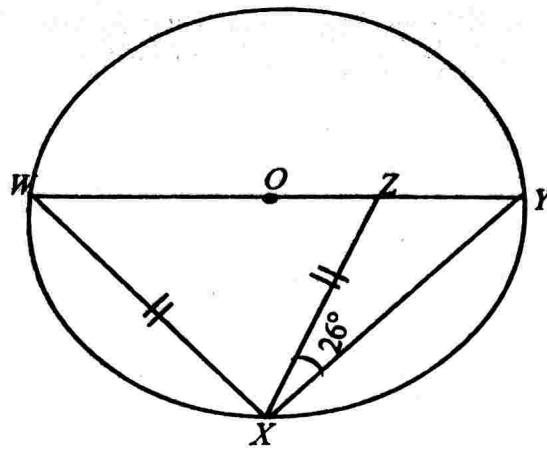
17. Given that $x^2 - 11x + m$ is a perfect square, find the value of m .

- A. $\frac{11}{2}$
- B. $\frac{121}{8}$
- C. $\frac{121}{4}$
- D. 121

18. If $x : y : z = 2 : 3 : 4$, evaluate $\frac{9x + 3y}{6z - 2y}$.

- A. 1.5
- B. 3.0
- C. 2.0
- D. 2.5

19.



NOT DRAWN TO SCALE

In the diagram, O is the centre of the circle WXY . $|WX| = |XZ|$ and $\angle ZXY = 26^\circ$. Find $\angle XYZ$.

- A. 62°
- B. 42°
- C. 32°
- D. 52°

20. Given that $\frac{16}{9}, x, 1, y$ is a Geometric Progression (G. P), find the value of xy .

- A. $\frac{9}{16}$
- B. 1
- C. $\frac{25}{12}$
- D. 2

21. A cylindrical container closed at both ends has radius 5 cm and height 10 cm. Calculate, correct to two decimal places, the total surface area. [Take $\pi = \frac{22}{7}$]

- A. 456.43 cm^2
- B. 460.43 cm^2
- C. 470.43 cm^2
- D. 471.43 cm^2

22. The height of a square base pyramid is **thrice** the length of a side of its base. If the base area is 324 cm^2 , find the volume of the pyramid.

A. $17,496 \text{ cm}^3$
 B. 972 cm^3
 C. $5,832 \text{ cm}^3$
 D. 324 cm^3

23. The sides of a scalene triangle are 4 cm, 9 cm and 11 cm. Calculate, correct to the **nearest** whole number, the area of the triangle.

A. 19 cm^2
 B. 21 cm^2
 C. 13 cm^2
 D. 17 cm^2

24. Determine the **least** value of x such that $7 + x = 3(\text{mod } 8)$.

A. 3
 B. 6
 C. 4
 D. 5

25. The angle of elevation of the top of a vertical pole from a point, P on a level ground is 60° . The distance from P to the foot of the pole is 55 m. Find the height of the pole.

A. $\frac{55\sqrt{3}}{2} \text{ m}$
 B. $\frac{55\sqrt{3}}{3} \text{ m}$
 C. $55\sqrt{3} \text{ m}$
 D. $\frac{55}{2} \text{ m}$

26. One side of a gutter is 15 cm lower than the other side. A plank of wood which is laid across the gutter to form a bridge, slopes at an angle of 35° to the horizontal. How wide is the gutter?

A. 10.50 cm
 B. 18.31 cm
 C. 21.42 cm
 D. 26.15 cm

27. The gradient of the line passing through the points (3, 6) and (x, 4) is $-\frac{2}{5}$. Find the value of x.
- A. 8
B. 6
C. 5
D. 3
28. If $P = \{x: 1 \leq x \leq 6\}$ and $Q = \{x: 2 < x < 9\}$ where $x \in R$, find $P \cap Q$.
- A. $\{x: 2 < x < 6\}$
B. $\{x: 2 \leq x < 6\}$
C. $\{x: 2 < x \leq 6\}$
D. $\{x: 2 \leq x \leq 6\}$
29. A student measured the length of a classroom and obtained 3.99 m which is less than the actual length. If the percentage error was 5 %, what was the actual length?
- A. 3.78 m
B. 3.80 m
C. 4.20 m
D. 4.18 m
30. Yakubu received $12\frac{1}{2}\%$ of the sales made in a certain month. If the amount he received was \$ 35,000.00, what was the total sales made?
- A. \$ 140,000.00
B. \$ 245,000.00
C. \$ 315,000.00
D. \$ 280,000.00
31. A woman pours 85 litres of kerosene into a cylindrical container with radius 7 cm. Calculate, correct to the nearest cm, the depth of the kerosene in the container. [Take $\pi = \frac{22}{7}$]
- A. 595 cm
B. 552 cm
C. 480 cm
D. 240 cm
32. The lines $3x + 2y = 4$ and $y = 2x - 5$ intersect at a point $P(x, y)$. Find the coordinates of P .
- A. (-2, 1)
B. (2, -1)
C. (-2, -1)
D. (2, 1)

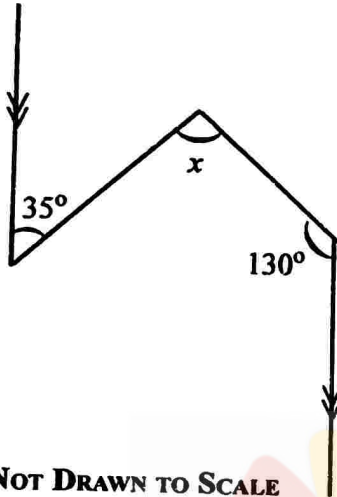
33. In the first year, Mr. Kwakye's annual salary was \$1,560.00. His salary was increased each year by a constant value, y until it was \$13,980.00 in the 13th year. Calculate the value of y .
- \$1,129.09
 - \$1,035.00
 - \$955.38
 - \$1,230.00
34. If $\log x = 2 - 3\log 2$, find the value of x .
- $\frac{2}{5}$
 - $\frac{25}{2}$
 - $\frac{2}{25}$
 - $\frac{5}{2}$
35. A woman bought a washing machine for \$18,000.00. If the exchange rate is \$0.045 to ₵1.00, find in ₵, the cost of the machine.
- ₵ 600,000.00
 - ₵ 250,000.00
 - ₵ 300,000.00
 - ₵ 400,000.00
36. Evaluate: $141_{\text{six}} + 233_{\text{six}} - 102_{\text{six}}$.
- 412_{six}
 - 405_{six}
 - 312_{six}
 - 305_{six}

37. Badu is four times as old as Juliet. In 10 years Badu will be twice as old as Juliet. Find Juliet's age.
- 6 years
 - 5 years
 - 3 years
 - 8 years
38. Make u the subject of the relation $\frac{t}{s+u} = \frac{s}{t-u}$.
- $u = t - s$
 - $u = ts$
 - $u = s - t$
 - $u = t + s$
39. A car covers the first 80 km of a journey in 2 hours and completes the journey by travelling further for 2.5 hours at 50 km/h. What is the average speed of the entire journey?
- $44 \frac{4}{9}$ km/h
 - $63 \frac{1}{4}$ km/h
 - $47 \frac{1}{4}$ km/h
 - $45 \frac{5}{9}$ km/h
40. Consider these two statements:
 P: N is an odd number
 Q: N is a prime number greater than 2.
 Express "If N is not an odd number then N is not a prime number greater than 2" in symbolic form.
- $\sim P \wedge \sim Q$
 - $\sim P \Rightarrow \sim Q$
 - $\sim P \Rightarrow Q$
 - $P \Rightarrow \sim Q$

41. A chord of a circle, 12 cm long subtends an angle of 150° at the centre of the circle. Find the radius of the circle.
- A. 3.0 cm
B. 4.3 cm
C. 6.0 cm
D. 6.2 cm
42. The graph of $y = x^2 - 5x + k$ passes through the point (3, 1). Find the value of k .
- A. 2
B. 3
C. 7
D. 5
43. Find the values of x for which $\frac{x+1}{3x^2-12}$ is not defined.
- A. $x = 0, 2$
B. $x = -2, 2$
C. $x = 4, -4$
D. $x = 2, -4$
44. If $4^x = \frac{1}{1024}$, find the value of x .
- A. -10
B. 8
C. 4
D. -5
45. A point on the ground is 5 m away from the foot of a vertical wall 12 m high. Calculate, correct to the nearest degree, the angle of depression of the point from the top of the wall.
- A. 25°
B. 67°
C. 65°
D. 23°

46. Find the mean of $(x + y)$, $(2x + 3y)$, $(2x - 2y)$ and $(3x - 2y)$.
- A. $2x$
 B. $4x$
 C. $3x$
 D. x

47.



Find the value of x in the diagram.

- A. 95°
 B. 165°
 C. 85°
 D. 50°

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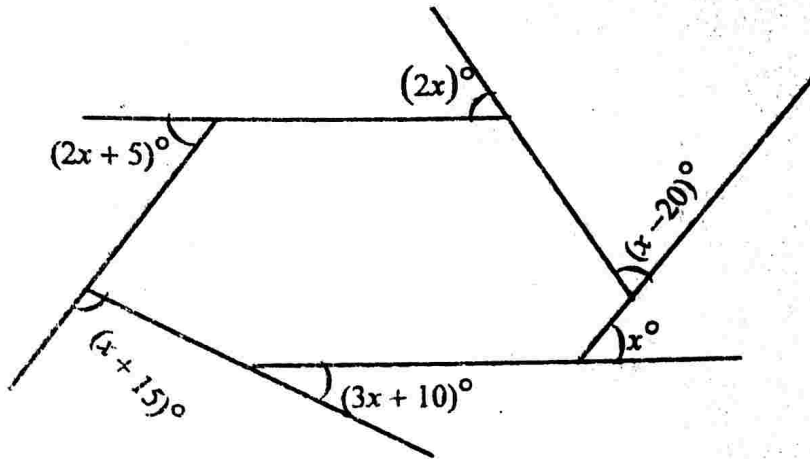
Number of subjects	1	2	3	4	5	6	7	8
Number of students	2	1	4	3	8	5	4	3

The table shows the number of subjects registered by a class of students preparing for an examination.

Use the information to answer questions 48 and 49.

48. Find the median.
- A. 2
 B. 5
 C. 8
 D. 3
49. Calculate the mean of the distribution.
- A. 3
 B. 5
 C. 8
 D. 2

50.



NOT DRAWN TO SCALE

Find the value of x in the diagram.

- A. 35
- B. 37
- C. 31
- D. 41